

CARE V5 & V6: Resolving the BatchNorm Paradox

A Rigorous Cross-Architecture Ablation Suite for NeuRIPS

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Recap: The V4 "BatchNorm Paradox"

The Challenge:

- In deep SNNs utilizing Batch Normalization (e.g., SEW-ResNets), CARE's homeostatic magnitude scaling of synaptic weights (ΔW) was effectively **nullified**.
- BN squashes the pre-activation variance back to $\sigma^2 = 1.0$, preventing the targeted rescue of dead/silent neurons.
- Instead of rescuing neurons, standard CARE acted as **unsupervised directional noise**, introducing a "Plasticity Tax" dropping CIFAR-100 accuracy by 14.7%.

The Mission: Definitively separate CARE's true rescue capability ("The Lazarus Effect") from the architectural interference of BatchNorm.

The New Methodology: Bypassing the Nullspace

Targeting the Affine Scale (γ):

- Instead of scaling the weights preceding the BN layer, CARE V5 shifts to targeting the explicit scale parameter (γ) of the BN layer itself.
- *Why?* γ operations occur **after** normalization, directly controlling the variance that reaches the spiking threshold without being cancelled out.

The NeurIPS V5 Ablation Axes

- ① **Homeostatic Target Modification:** Weight vs. Gamma (γ) vs. Both.
- ② **SNR / Grace Period:** Isolating stable backpropagation phases from dynamic interventions.
- ③ **Learning Rate Sensitivity (η):** Finding the stability threshold.
- ④ **Dataset Scaling:** Moving to CIFAR-100 to prove generalization.

Expanding the Scope: The V6 Cross-Architecture Suite

Supervisor Feedback constraint: "ResNet metrics alone might be deemed cliché or overly constrained."

Solution: Create a universal standard validating CARE across distinct topological paradigms capable of isolating or amplifying the "Lazarus Effect."

- **Plain ConvNets:** No skip connections, no BN. The ultimate "vanilla" testbed where standard CARE shines strongest.
- **VGG-SNNs:** Deep, chain-like feedforward paths prone to catastrophic starvation.
- **SEW-ResNets:** Our robust baseline, equipped with Gamma-targeted CARE.
- **Spiking Attention (Transformer-style):** Exploring CARE's stability on sequence-to-sequence spatial architectures.

The Ultimate Test: The "Sabotage / Lazarus" Initialization

Proof of Resilience:

- A proper model must not only perform well but reliably recover from adversarial or catastrophic initial states.
- We introduce the **Sabotage Initialization**: Forcing a massive proportion of network parameters into deeply suppressed/silent states across VGG, Plain, and ResNet architectures.

Architecture	Rescue Target	Normal Init	Sabotage Init
Plain ConvNet	Weight Magnitude	Expected ++	Essential
VGG-11	Gamma Scaling	Expected ++	Essential
Spiking Attention	Gamma Scaling	Unknown	Essential

Table: The V6 Architecture Test matrix

What these suites provide:

- ① **Table 1 - Cross-Architecture Rescue:** Demonstrating CARE is a universal solution for deep chain architectures, while safe for ResNets.
- ② **The BN Mathematical Proof:** A formal analysis demonstrating how classic STDP variants are incompatible with Modern DL normalization without Gamma targeting.
- ③ **The Lazarus Paradigm:** Setting a new standard in Neuromorphic engineering: Networks that dynamically **heal** their own computational graph during training.

Current Status: Experiments are queued locally and pending full GPU VRAM allocation.